Design Proposal: Unbalanced Mach-Zehnder Interferometers for Extracting the Group Index of a Silicon Strip Waveguide

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***Abstract*—This proposal describes three unbalanced Mach-Zehnder interferometers (MZIs) on a 220 nm silicon-on-insulator platform, designed to extract the group index n_g of a 500 nm wide strip waveguide (quasi-TE) from the free spectral range (FSR) of the measured transmission spectrum. The waveguide mode is simulated with the finite-difference eigenmode solver of Lumerical MODE. A second-order polynomial compact model of the effective index is fitted to the simulation, giving $n_{eff} = 2.4432$ and $n_g = 4.197$ at 1550 nm. The compact model is used to compute the MZI spectra, and the equation $n_g = \lambda_c^2/(\mathrm{FSR}\cdot\Delta L)$ is derived and tested on simulated spectra with path-length differences from 50 to 400 μm. Three designs ($\Delta L = 50$, 100 and 200 μm) are included in the layout, which measures 541 μm $\times$ 408 μm and passed the design-rule check and the SiEPIC-Tools verification without errors.**

***Index Terms*—silicon photonics, Mach-Zehnder interferometer, group index, free spectral range, waveguide dispersion.**

I. Introduction

Silicon-on-insulator (SOI) waveguides confine light strongly because of the large index contrast between the silicon core and the oxide cladding. The same strong confinement makes the dispersion of the guided mode depend on the waveguide geometry, so the group index of a fabricated waveguide can differ from the simulated one. Unbalanced MZIs are widely used in silicon photonics, for example as wavelength filters and as building blocks of modulators and delay lines [1], and they also turn the group index into something that is easy to measure: the spacing of the transmission fringes.

The design objective is a set of MZI circuits whose fringe spacing (FSR) is smaller than the roughly 50 nm measurement bandwidth, so that several fringes are visible. After fabrication and measurement, the extracted n_g will be compared with the simulated value reported here. In this proposal only the default strip waveguide (500 nm $\times$ 220 nm, quasi-TE) is simulated, and the design variations are the path-length difference ΔL.

II. Theory

An unbalanced MZI splits the input light into two arms with lengths L_1 and L_2, and recombines them. The path-length difference is $\Delta L = L_2 - L_1$. For ideal 50:50 splitters without loss, the phase difference between the two arms is

$$\Delta\phi(\lambda) = \frac{2\pi\, n_{eff}(\lambda)\,\Delta L}{\lambda}, \tag{1}$$

and the transfer function (output power over input power) is

$$T(\lambda) = \cos^2\!\left(\frac{\Delta\phi(\lambda)}{2}\right) = \frac{1}{2}\left[1 + \cos\!\left(\frac{2\pi\, n_{eff}(\lambda)\,\Delta L}{\lambda}\right)\right]. \tag{2}$$

The peaks of T occur where $\Delta\phi = 2\pi m$ with m an integer. The FSR is the wavelength spacing between two adjacent peaks, so $\Delta\phi$ changes by 2π across one FSR.

A. Derivation of the group index equation

Differentiating the phase with respect to wavelength, and keeping the wavelength dependence of n_{eff},

$$\begin{aligned}\frac{d\Delta\phi}{d\lambda} &= 2\pi\Delta L\left[\frac{1}{\lambda}\frac{dn_{eff}}{d\lambda} - \frac{n_{eff}}{\lambda^2}\right] \\ &= -\frac{2\pi\Delta L}{\lambda^2}\left[n_{eff} - \lambda\frac{dn_{eff}}{d\lambda}\right].\end{aligned} \tag{3}$$

The bracket is the group index,

$$n_g = n_{eff} - \lambda\frac{dn_{eff}}{d\lambda}, \tag{4}$$

so that $|d\Delta\phi/d\lambda| = 2\pi n_g \Delta L/\lambda^2$. Over one FSR the phase changes by 2π. To first order, i.e. treating $d\Delta\phi/d\lambda$ as constant over one FSR (valid when $\mathrm{FSR} \ll \lambda$),

$$\frac{2\pi\, n_g\,\Delta L}{\lambda^2}\,\mathrm{FSR} = 2\pi \;\;\Rightarrow\;\; \mathrm{FSR} = \frac{\lambda^2}{n_g\,\Delta L}, \tag{5}$$

$$n_g = \frac{\lambda_c^2}{\mathrm{FSR}\cdot\Delta L}, \tag{6}$$

where λ_c is the centre wavelength of the two peaks used to measure the FSR. The phase itself depends on n_{eff}, while the fringe spacing depends on n_g; this is why the measurement gives n_g and not n_{eff}.

III. Modelling and Simulation

A. Waveguide geometry and mode

The waveguide is a silicon strip, 500 nm wide and 220 nm high, surrounded by SiO_2, and the mode is the fundamental quasi-TE mode. It was solved with the finite-difference eigenmode (FDE) solver of Lumerical MODE, with perfectly matched layer boundaries and a 10 nm mesh around the core (20 nm elsewhere). At 1550 nm the mode has a TE polarization fraction of 0.984, and $n_{eff} = 2.4432$. Fig. 1 shows the normalised field magnitude; the field is concentrated in the core and decays into the cladding.

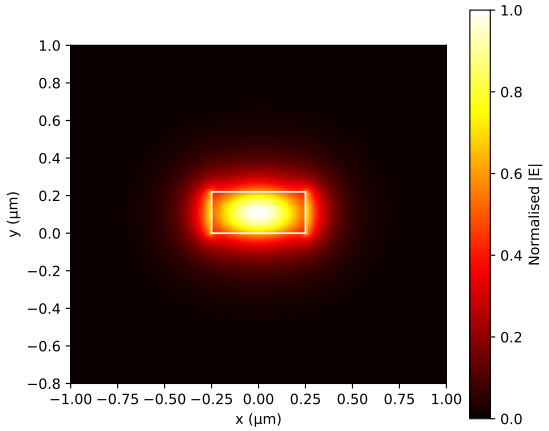


Fig. 1. Normalised $|E|$ of the fundamental quasi-TE mode of the 500 nm × 220 nm strip waveguide at 1550 nm (Lumerical MODE FDE). The white rectangle is the silicon core.

B. Circuit geometry

Each circuit (Fig. 2) is an unbalanced MZI. Light is coupled in and out of the chip with fibre grating couplers (GC), a Y-branch splits it into two arms, and a second Y-branch recombines it. The two arms use the same waveguide, and differ only in length: $L_2 = L_1 + \Delta L$. The three designs share this layout and use the values of ΔL in Table I. The circuits were generated with the gdsfactory MZI component and the `ebeam_y_1550` Y-branch of the UBC PDK, and the arm-length difference is set by the `delta_length` parameter of that component.

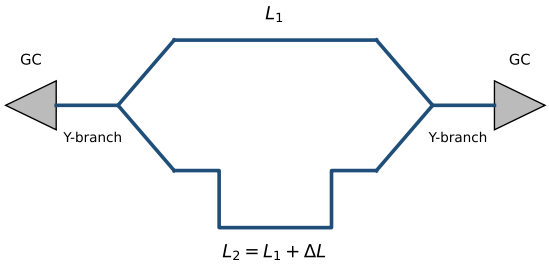


Fig. 2. Schematic of the unbalanced MZI (not to scale): input and output grating couplers (GC), two Y-branches, a short arm of length L_1 and a long arm of length $L_2 = L_1 + \Delta L$.

C. Effective and group index

n_{eff} was computed from 1500 to 1600 nm in 5 nm steps, and n_g was obtained from Eq. (4) by finite differences. Fig. 3 shows both. n_{eff} decreases almost linearly with wavelength, while n_g varies only slightly, from 4.206 at 1500 nm to 4.191 at 1600 nm.

A numerical issue was found and corrected. The default tabulated silicon data in Lumerical (Palik) has only a few points near 1.5 μm (at 1.4, 1.532, 1.6 and 1.696 μm), and the solver interpolates it linearly. This produced a kink in $n(\lambda)$ at 1.532 μm and an artificial step of about 0.09 in the simulated n_g around 1530–1540 nm. The step disappeared after the Si and SiO_2 data were fitted smoothly and resampled on a dense wavelength grid (601 points between 1.4 and 1.7 μm), and all results in this report use the resampled materials.

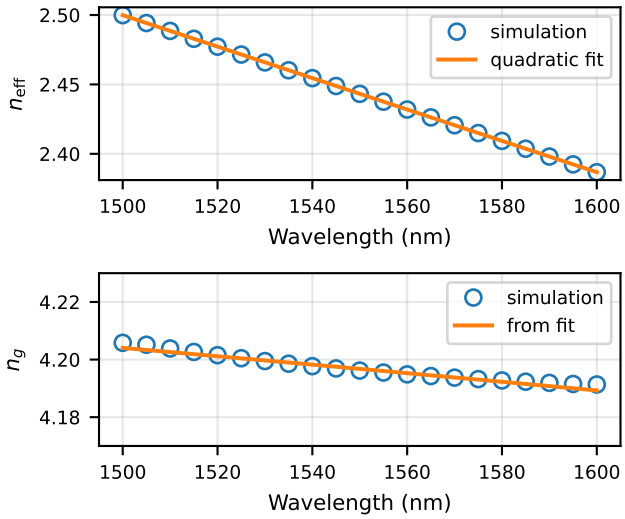


Fig. 3. Effective index (top) and group index (bottom) of the quasi-TE mode versus wavelength. Markers: simulation. Lines: second-order compact model, and the group index computed from it.

D. Compact model

A least-squares fit of n_{eff} around $\lambda_0 = 1.55$ μm gives

$$n_{eff}(\lambda) = 2.44321 - 1.13134\,(\lambda - \lambda_0) + 0.04769\,(\lambda - \lambda_0)^2, \tag{7}$$

with λ in μm. The maximum fit error of n_{eff} is 1.1×10^{-5}. Using Eq. (4), the compact model gives

$$n_g(\lambda) = n_{eff}(\lambda) - \lambda\,[-1.13134 + 2(0.04769)(\lambda - \lambda_0)]\,, \tag{8}$$

so $n_g(\lambda_0) = 2.44321 + 1.55 \times 1.13134 = 4.197$. The simulated value at 1550 nm is 4.196, and the largest difference between the simulated n_g and the model over 1500–1600 nm is 0.002.

E. MZI spectrum and parameter variations

Inserting Eq. (7) in Eq. (2) gives the ideal MZI spectrum. Fig. 4 shows the case $\Delta L = 100$ μm, with an FSR of about 5.7 nm.

Three designs are included in the layout, all with the 500 nm wide quasi-TE strip waveguide and Y-branch splitters, differing in ΔL (Table I). The FSR range of 2.9–11.4 nm is well below the 50 nm measurement bandwidth, so several fringes are expected in the measured band; whether the smallest FSR (2.9 nm) is resolved well enough depends on the wavelength step of the measurement system, which is to be confirmed. For each design the FSR was measured from the simulated spectrum (mean spacing of the peaks between 1500 and 1600 nm), and n_g was recomputed with Eq. (6) and compared with the compact-model value at λ_c.

FSR_{pred} is $\lambda_0^2/(n_g(\lambda_0)\Delta L)$ with $n_g(\lambda_0) = 4.197$. The extracted n_g agrees with the model within 0.04% for the three

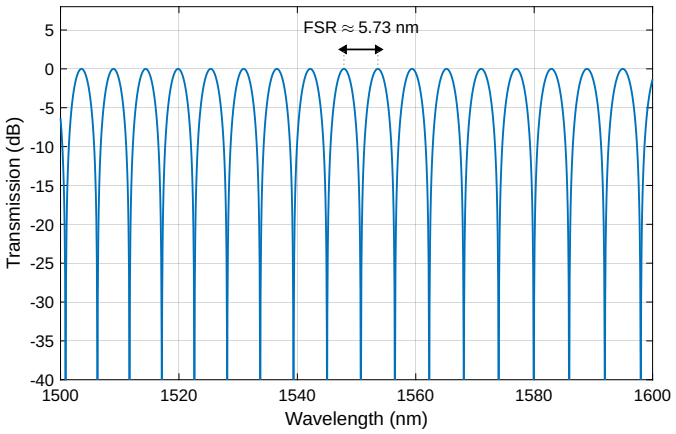


Fig. 4. Transmission of the MZI with $\Delta L = 100$ μm, computed from the compact model with ideal 50:50 splitters and no loss. Values below -40 dB are cut off.

TABLE I
MZI DESIGNS (500 NM WIDE, QUASI-TE) AND TEST OF EQ. (6) ON THE SIMULATED SPECTRA.

ID	ΔL (μm)	FSR$_{pred}$ (nm)	FSR$_{meas}$ (nm)	n_g extracted	n_g model	Error (%)
1	50	11.449	11.428	4.1962	4.1970	-0.019
2	100	5.725	5.714	4.1959	4.1970	-0.026
3	200	2.862	2.857	4.1956	4.1970	-0.034

designs, which supports Eq. (6). The same test was repeated for $\Delta L = 150$, 300 and 400 μm, which are not in the layout, and gave errors of -0.032, -0.035 and -0.036%. The small differences between FSR$_{pred}$ and FSR$_{meas}$ are consistent with the wavelength dependence of λ^2/n_g across the 100 nm band, since FSR$_{pred}$ uses n_g at 1550 nm only while FSR$_{meas}$ is averaged over the whole band.

F. Extraction from measured data

From a measured spectrum, the positions of at least two adjacent peaks (or notches) are read, their spacing gives the FSR and their mean wavelength gives λ_c. Then $n_g = \lambda_c^2/(\mathrm{FSR} \cdot \Delta L)$ with the designed ΔL. Repeating this over several peaks and over the three designs gives n_g at different wavelengths and a check of the $1/\Delta L$ dependence of the FSR. The extracted value is the group index at λ_c, not at 1550 nm in general.

G. Assumptions and limits

The model assumes ideal 50:50 splitters and no propagation loss, so the simulated peaks reach 0 dB and the notches are deep; in measurements the peak level will be lower and the extinction ratio finite. Only the quasi-TE mode of the 500 nm wide waveguide has been simulated. Other widths and the quasi-TM mode are not included in this proposal.

IV. FABRICATION LAYOUT

The layout (Fig. 5) was generated with gdsfactory and the UBC PDK (`ubcpdk`) for the EBeam process, using the `ebeam_y_1550` Y-branch [2] and the `ebeam_gc_te1550` grating coupler [3]. It contains the three MZIs of Table I side by side, all with a 500 nm wide strip waveguide for the quasi-TE mode at 1550 nm.

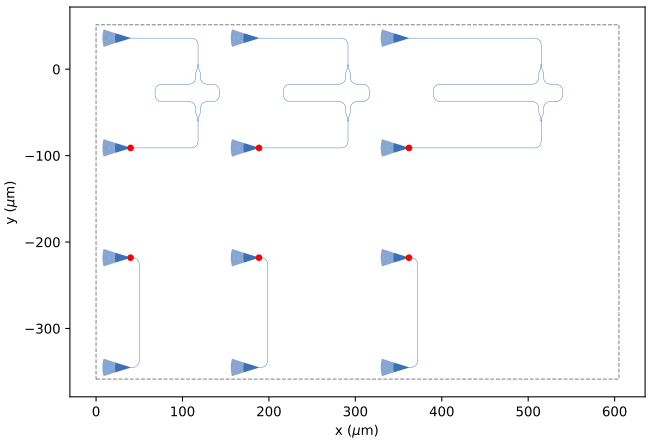


Fig. 5. Layout of the three MZIs ($\Delta L = 50$, 100 and 200 μm, from left to right). Blue: silicon (layer 1/0); green outlines: device boundaries; red dots: measurement labels; dashed rectangle: floor plan (layer 99/0). In each fibre array the MZI input (labeled) is the second grating coupler from the top, and the reference loopback is formed by the two lower grating couplers.

Each MZI has its own fibre array of four grating couplers at a vertical pitch of 127 μm. From top to bottom they are: the MZI output, the MZI input, the reference loopback input and the reference loopback output. The two reference couplers are joined by a loopback waveguide, which serves as a de-embedding structure for the insertion loss of the measurement system; the output is one pitch below the labeled input. The MZI measurement labels are `opt_in_TE_1550_device_wuweichen-MZI.dL50`, `...dL100` and `...dL200`, on layer 10/0 at the second coupler from the top, so that the MZI output detector is one pitch above the labeled input. The loopbacks have their own labels, `...-REF.dL50`, `...-REF.dL100` and `...-REF.dL200`. The six labels are all different.

The layout is 541 μm $\times$ 408 μm, inside the 605 μm $\times$ 410 μm floor plan (drawn on layer 99/0), with a silicon-to-silicon distance of about 13 μm between neighbouring designs. The width of one design grows with ΔL (144, 169 and 219 μm for $\Delta L = 50$, 100 and 200 μm), because the longer arm extends sideways by about $\Delta L/2$. The longest design that fits in the floor plan is $\Delta L = 200$ μm, so the designs with $\Delta L = 300$ and 400 μm considered in the simulations could not be included.

The layout was checked with SiEPIC-Tools v0.5.31 in KLayout v0.30.12 with the EBeam technology v0.4.53, which reported 0 errors, and the design-rule check (minimum feature size 60 nm) also reported 0 errors. The features of the design review checklist that were verified directly in the GDS file are: the grating-coupler pitch of 127 μm, the position and uniqueness of the labels, the presence of the loopback structure, the layers used, the file name `EBeam_wuweichen.gds` and the top-cell name `EBeam_wuweichen`.

The layout was also run through the course's automated circuit simulation, which uses the netlist extracted by SiEPIC-Tools and the compact models of the EBeam PDK (Fig. 6). The three MZIs show fringes whose spacing decreases with ΔL, and the reference loopback shows the smooth response of two grating couplers and a waveguide. The FSR near 1550 nm was read from the notch positions in the plots (Table II); the reading accuracy is about ± 0.2 nm for $\Delta L = 50$ μm and ± 0.1 nm for the other two. With $n_g = \lambda^2/(\mathrm{FSR}\,\Delta L)$ and $\lambda = 1550$ nm, the three designs give $n_g \approx 4.18$, which agrees with the compact-model value 4.197 within the reading accuracy. This only shows that the layout implements the intended ΔL (the extracted netlist gives arm lengths that differ by exactly 50.0 μm for the first design); it is not a measurement, because the simulation uses the same compact model. The peak transmission of the MZIs (≈ -5 dB) is close to that of the reference loopback (≈ -4.8 dB), so the grating couplers account for almost all of the loss in this model. The three reference loopbacks are identical and gave the same curve.

TABLE II
FSR AND GROUP INDEX READ FROM THE LAYOUT SIMULATION (FIG. 6) NEAR 1550 NM.

ΔL (μm)	FSR$_{pred}$ (nm)	FSR$_{read}$ (nm)	n_g
50	11.45	11.5	4.18
100	5.72	5.75	4.18
200	2.86	2.87	4.18

V. EXPERIMENTAL DATA

To be completed after measurement.

VI. ANALYSIS

To be completed after measurement.

VII. CONCLUSION

To be completed at the end of the course.

REFERENCES

[1] L. Chrostowski and M. Hochberg, *Silicon Photonics Design: From Devices to Systems*. Cambridge University Press, 2015.

[2] Y. Zhang *et al.*, *Opt. Express*, vol. 21, no. 1, 2013 (Y-branch, as referenced in the `ebeam_y_1550` component of the EBeam PDK).

[3] Y. Wang *et al.*, *Opt. Express*, vol. 22, 2014, doi:10.1364/OE.22.020652 (grating coupler, as referenced in the `ebeam_gc_te1550` component of the EBeam PDK).

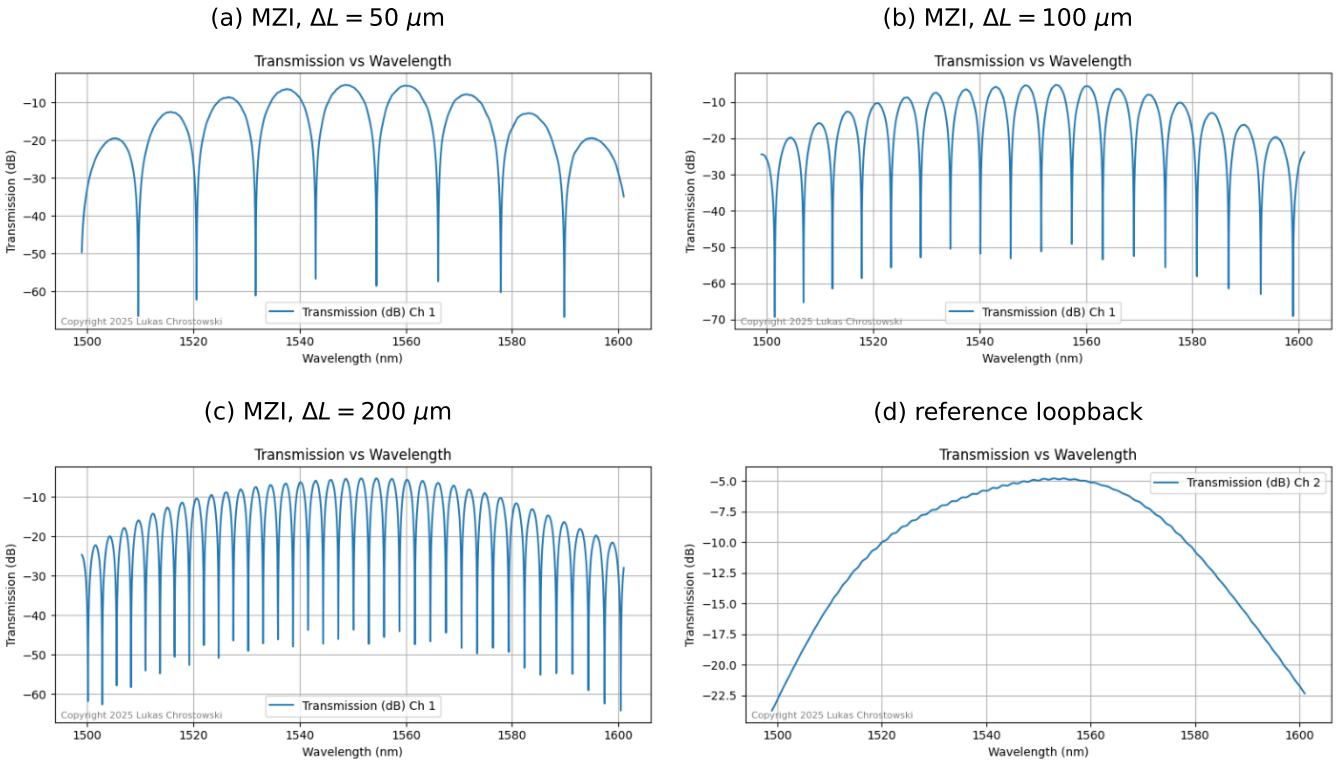


Fig. 6. Circuit simulation of the submitted layout (course server, compact models). (a)–(c) MZIs with $\Delta L = 50$, 100 and 200 μm; (d) reference loopback (the three loopbacks give the same curve).